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PS No: 21 | **Title:** Kidney Abnormality Detection from CT
GitHub (notebook): [add repository link here]

1. Problem and Objective

Given a single abdominal CT slice, the task is to classify it as Normal, Cyst, Stone or Tumor. Cysts, stones and tumors need very different management, and a missed tumor or stone delays treatment while specialist radiologists are scarce, so a reliable first-pass classifier is useful. Because the classes are imbalanced (Normal is 3.69 times larger than Stone), plain accuracy can hide weak minority-class performance. Success is therefore judged on the held-out **test set** by **macro F1**, with per-class recall for Stone (the smallest class) and Tumor (the most serious class) as the key checks.

2. Dataset and Preprocessing

Source: Kaggle "CT KIDNEY DATASET: Normal-Cyst-Tumor and Stone" (Islam et al., 2022; link at the end). The **full set of 12,446 images** was used: Normal 5,077, Cyst 3,709, Tumor 2,283, Stone 1,377 (no subset). EDA consisted of class counts and one sample image per class. **Split:** stratified random 70/15/15 with seed 42, giving 8,712 train / 1,867 validation / 1,867 test images; the test split was used once, for the final evaluation only. **Preprocessing:** each image is decoded as 3-channel, resized to 224x224 and scaled with MobileNetV2's *preprocess_input* (pixel range -1 to 1). **No augmentation** was applied. **Imbalance:** balanced class weights in the loss (Cyst 0.84, Normal 0.61, Stone 2.26, Tumor 1.36). **Dataset risk:** file names carry no patient ID, so the split is image-level and slices from one patient may fall in both train and test (possible leakage); duplicates were not checked.

3. Methodology

Transfer learning: MobileNetV2 pretrained on ImageNet is used as a **fully frozen** feature extractor (2,257,984 frozen parameters), followed by global average pooling, dropout (0.3) and a 4-way softmax layer. Only the 5,124 parameters of the final dense layer are trained. MobileNetV2 was chosen because it is lightweight (8.63 MB), trains in minutes and can run on modest hospital hardware. Training used Adam with weighted sparse categorical cross-entropy; early stopping on validation loss (patience 2, best weights restored) and learning-rate reduction on plateau (factor 0.5, patience 1) were enabled, but neither triggered because validation loss fell in every epoch. Environment: TensorFlow 2.21.0, Google Colab GPU. Seed 42 fixes the split and the shuffle buffer; a global framework seed was not set, so the exact numbers can vary slightly between runs.

Table 1. Training configuration.

Hyperparameter	Value	Hyperparameter	Value
Backbone	MobileNetV2 (ImageNet), frozen	Optimiser / initial LR	Adam / 0.001
Input size / batch size	224x224x3 / 32	Loss	Class-weighted sparse CE
Head	GAP, Dropout 0.3, Dense(4)	Max epochs / epochs run	5 / 5
Trainable parameters	5,124 of 2,263,108	Callbacks	EarlyStopping, ReduceLROnPlateau

4. Results

All numbers below are from the **test split** (n = 1,867) unless stated otherwise; per-class cells show precision / recall / F1.

Table 2. Test-set results. Train time is the sum of the five logged epoch times (Colab GPU).

Model	Test acc.	Macro F1	Cyst	Normal	Stone	Tumor	Train time
MobileNetV2, frozen + class weights	95.66%	0.94	0.98/0.96/0.97	0.99/0.96/0.97	0.88/0.92/0.90	0.91/0.96/0.93	199 s

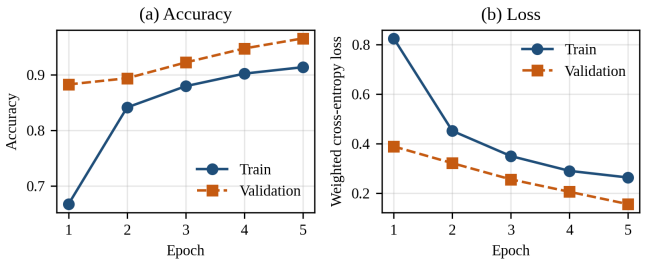


Fig. 1. Learning curves: training (running mean, dropout on) vs validation.

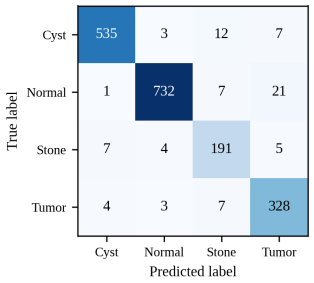


Fig. 2. Test-set confusion matrix (counts).

5. Analysis and Discussion

Confusions. The model made 81 errors on 1,867 test images (Fig. 2). The largest single confusion is Normal predicted as Tumor (21 images, 25.93% of all errors), followed by Cyst predicted as Stone (12). Stone has the lowest precision (0.88): 26 non-stone images were labelled Stone (12 Cyst, 7 Normal, 7 Tumor), against 16 true stones missed. Only 3 of 342 tumors (0.88%) were called Normal, so the errors lean towards false alarms rather than missed tumors. A likely reason, which was not tested, is that a frozen ImageNet backbone sees a whole-abdomen slice shrunk to 224x224, so small stones and subtle soft-tissue differences carry few pixels. **Fitting.** There is no sign of over-fitting: validation accuracy (96.57%) exceeds the final running training accuracy (91.38%, dropout active) and validation loss was still falling at epoch 5 (0.1554), so the model is probably under-trained. Test accuracy (95.66%) is 0.91 points below validation accuracy. **Misclassified examples.** The first four errors inspected (three Cyst predicted as Tumor, one Normal predicted as Stone) are full abdominal slices in both axial and coronal views where the kidney fills a small part of the frame. **Specific challenge.** Class weights (Stone weight 2.26) were used for the minority class and give Stone recall of 92.27% (191 of 207), higher than its precision; an unweighted run was not made, so the weights' exact effect is not isolated.

6. Real-world Relevance, Limitations and Ethics

Such a model could act as a triage or second-reader tool that flags abnormal kidney CT slices where radiologists are scarce. It is not deployable today: it was tested on one public dataset with a probable patient-level leakage risk, has no external validation, works on single slices rather than whole scans, and gives no localisation or uncertainty to justify a prediction. Scanner, age and sex information is absent, so bias across groups cannot be audited, and the de-identification of the source images must be confirmed before any clinical use. A missed tumor delays treatment and a false alarm (21 Normal images flagged as Tumor here) causes anxiety and unnecessary work-up, so output must support, not replace, a radiologist.

7. Conclusion and Future Work

A frozen MobileNetV2 with class weights reached 95.66% test accuracy, macro F1 of 0.94 and Stone recall of 92.27% after only 199 s of training. Next, I would split by patient (if IDs can be obtained), unfreeze the top MobileNetV2 blocks at a low learning rate with augmentation, and check whether Stone recall and the Normal-to-Tumor errors improve on the same test split.

Dataset: Islam, M. N. et al., "Vision transformer and explainable transfer learning models for auto detection of kidney cyst, stone and tumor from CT-radiography," *Scientific Reports* 12, 11440 (2022), doi:10.1038/s41598-022-15634-4; data: kaggle.com/datasets/nazmul0087/ct-kidney-dataset-normal-cyst-tumor-and-stone.